Probability

Contents

- Probability
- Conditional Probability
- Bayesian Reasoning
- Independence

Probability (1)

- Data analysis is strongly related to probability
- Because of its firm mathematical foundation, many ML approaches have been developed based on probability.
- Naïve Bayesian Model, Bayesian Network, Hidden Markov Model, etc.

Probability (2)

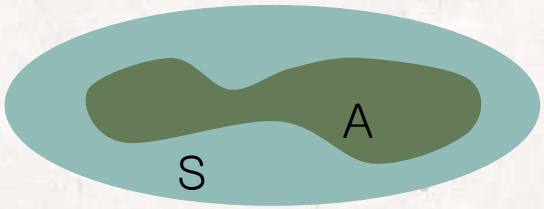
- Properties
  - If $A \subseteq B \subseteq U$, then

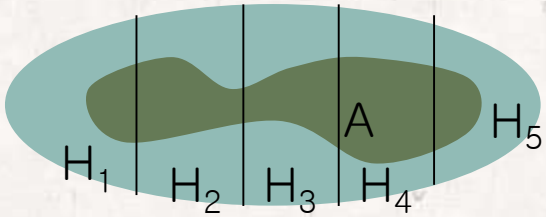
    $0 \le P(A) \le P(B) \le 1$
  - If $A, B \subseteq U$, then

    $P(A,B) + P(A,\neg B) = P(A)$
  - If $A, B \subseteq U$, then

    $P(A \text{ or } B) = P(A) + P(B) - P(A,B)$
  - If $H_i \subseteq U$ for $1 \le i \le n$, $H_i \cap H_j = \varnothing$ whenever $i \neq j$ and $H_1 \cup H_2 \cup \ldots \cup H_n = U$, then

    $P(A) = P(A,H_1) + P(A,H_2) + \ldots + P(A,H_n)$



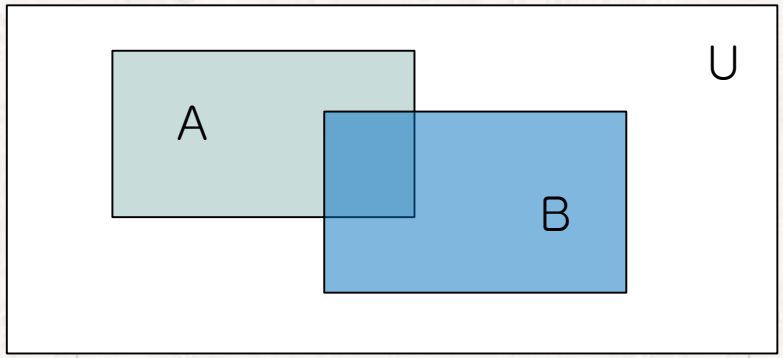


Conditional Probability (1)

- Definition

$$P(A|B) = \frac{P(A,B)}{P(B)}$$

  - Meaning
    - The probability that A will happen under the assumption that B already happened
    - The probability of A under the assumption that B is the universal set.



Conditional Probability (2)

- Of which followings can be defined?
  - A is a cause of B. We know that A happened.
    - P(A), P(B), P(A|B), P(B|A)
  - A is a cause of B. We know that A did not happen.
    - P(A), P(B), P(A|B), P(B|A)
  - A is a cause of B. We do not know whether A happened.
    - P(A), P(B), P(A|B), P(B|A)
  - A is a cause of B. We know that B happened.
    - P(A), P(B), P(A|B), P(B|A)
  - A is a cause of B. We know that B did not happen.
    - P(A), P(B), P(A|B), P(B|A)
  - A is a cause of B. We do not know whether B happened.
    - P(A), P(B), P(A|B), P(B|A)
  - We do not know the causal relation between A and B. We know that A happened.
    - P(A), P(B), P(A|B), P(B|A)
  - We do not know the causal relation between A and B. We know that A did not happen.
    - P(A), P(B), P(A|B), P(B|A)
  - We do not know the causal relation between A and B. We do not know whether A happened.
    - P(A), P(B), P(A|B), P(B|A)

Conditional Probability (3)

- Properties
  - Variations

$$P(A|B,C) = \frac{P(A,B,C)}{P(B,C)} \qquad P(A|B,C) = \frac{P(A,B|C)}{P(B|C)}$$

  - If $A \subseteq B \subseteq U$, then

$$0 \le P(A|C) \le P(B|C) \le 1$$

  - If $A, B \subseteq U$, then

$$P(A,B|C) + P(A,\neg B|C) = P(A|C)$$

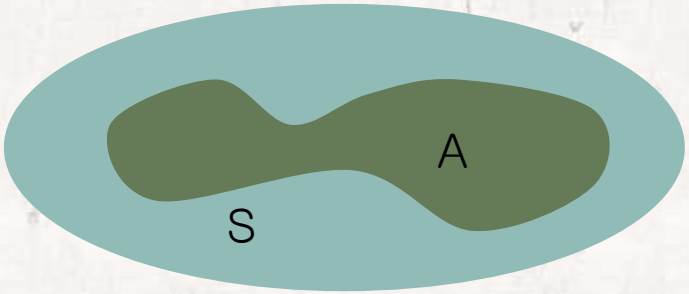
  - If $A, B \subseteq U$, then

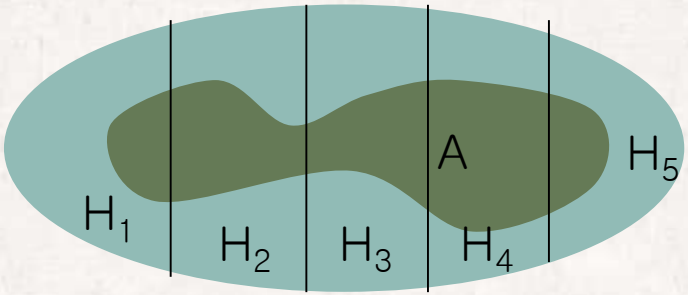
$$P(A \text{ or } B|C) = P(A|C) + P(B|C) - P(A,B|C)$$

Conditional Probability (4)

- Properties –con'd
  - If $H_i \subseteq S$ for $1 \le i \le n$, $H_i \cap H_j = \varnothing$ whenever $i \neq j$ and $H_1 \cup H_2 \cup \ldots \cup H_n = S$, then

$$P(A) = P(A,H_1) + P(A,H_2) + \ldots + P(A,H_n)$$
$$= P(A|H_1)P(H_1) + P(A|H_2)\,P(H_2) + \ldots + P(A|H_n)\,P(H_n)$$





Conditional Probability (5)

- Chaining

$$P(A,B,C,D) = P(A|B,C,D)P(B|C,D)P(C|D)P(D)$$

$P(A,B,C,D) = P(A|B,C,D)*P(B,C,D)$

$P(B,C,D) = P(B|C,D)*P(C,D)$

$P(C,D) = P(C|D)*P(D)$

  - Order is not important

$$\begin{aligned} P(A,B,C,D) &= P(A|B,C,D)P(B|C,D)P(C|D)P(D) \\ &= P(B|C,A,D)P(C|A,D)P(C|A)P(D) \\ &= P(A|D,C,B)P(D|C,B)P(D|C)P(B) \\ &\cdots \end{aligned}$$

$P(A,B,C,D) = P(B,C,A,D) = P(A,D,C,B) = \ldots$

Conditional Probability (6)

- Bayesian Rule
  - For B, $A \subseteq S$

Likelihood of B given A: L(B|A)

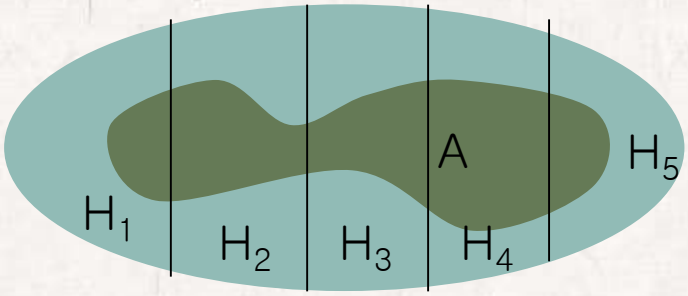
$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

  - Why important?
    - Usually, prior probabilities are easy to obtain.
    - Usually, one of P(A|B) or P(B|A) are easy to obtain.

=> Evaluating probability hard to obtain from ones easy to obtain

Conditional Probability (7)

- Bayesian Reasoning
  - If $H_1 \cup H_2 \cup \ldots \cup H_n = S$ and $H_i \cap H_j = \varnothing$ whenever $i \neq j$



$$P(H_i|A) = \frac{P(A|H_i)P(H_i)}{P(A|H_1)P(H_1) + P(A|H_2)\,P(H_2) + \ldots + P(A|H_n)\,P(H_n)}$$

Conditional Probability (8)

- Prior probability vs. Posterior probability
  - Prior probability
    - P(event)
    - Unconditioned probability of an event
    - Probability of an event prior any new evidence
      - Probability of an event without any information (knowledge)
  - Posterior probability
    - P(event | evidence)
    - Conditional probability of an event
    - Probability of an event given some new evidence
      - Probability of an event with some piece of information (knowledge)

Conditional Probability (9)

- Example: Bayesian Reasoning
  - We have three coins in a pockets, and choose one randomly and toss it
    - C_1: two heads
    - C_2: one head and one tail
    - C_3: two tails
  - What is the probability of each hypothesis?
    - $P(C_1)$, $P(C_2)$, $P(C_3)$
  - What is the probability of head and tail for each coin?
    - $P(h|C_1)$, $P(t|C_1)$ $P(h|C_2)$, $P(t|C_2)$ $P(h|C_3)$, $P(t|C_3)$
  - Now, we see a head but don't know which coin you chose. What is the probability of each coin?
    - $P(C_1|h)$, $P(C_2|h)$, $P(C_3|h)$

Bayesian Reasoning (1)

- Example
  - There is a disease that one person in 1,000 has
  - There is a screening test for the disease, that the test is always positive if the person has the disease,
  - and that it is also positive five percent of the time when the person does not have the disease
  - If a person tests positive, does he has the disease or not?

H_D : the people who have the disease
H_N : the people who do not have the disease
P : the people who test positive for the disease
N : the people who test negative for the disease

$P(H_D) = 0.001$
$P(H_N) = 0.999$
$P(P|H_D) = 1$
$P(P|H_N) = 0.05$

$P(H_D|P)$???

Bayesian Reasoning (2)

- Example –con'd

$$P(H_D|P) = \frac{P(P|H_D)P(H_D)}{P(P)}$$

$P(H_D) = 0.001$
$P(H_N) = 0.999$
$P(P|H_D) = 1$
$P(P|H_N) = 0.05$

$$P(P) = P(P|H_D)P(H_D) + P(P|H_N)P(H_N) \qquad D \cap H = \varnothing$$

$$P(H_D|P) = \frac{P(P|H_D)P(H_D)}{P(P)} = \frac{0.001}{0.05005} \approx 0.02$$

Independence (1)

- Definition
  - A is independent from B iff

$$P(A,B) = P(A)P(B)$$

$$P(A|B) = P(A|\neg B) = P(A)$$

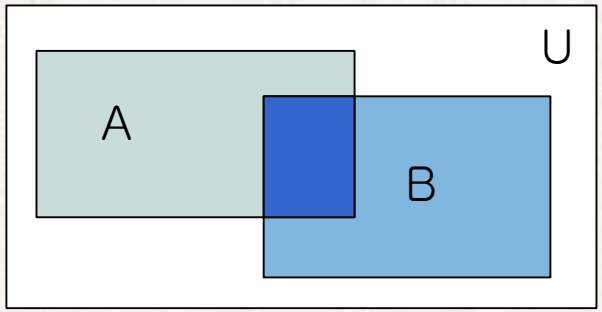
  - A and C are mutually independent given B iff

$$P(A,C|B) = P(A|B)P(C|B)$$

$$P(A|B,C) = P(A|B)$$

Independence (2)

- Some misunderstanding
  - A and B are independent, then $A \cap B = \varnothing$: right?

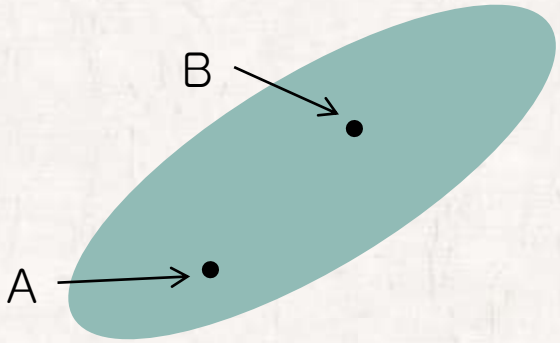


  - Which of these are true?
    - P(A,B) = P(A)P(B) ⇨ P(A,B|C)=P(A|C)P(B|C) for any C
    - P(A,B|C)=P(A|C)P(B|C) for given C ⇨ P(A,B) = P(A)P(B)
    - P(A,B|C)=P(A|C)P(B|C) for given C ⇨ P(A,B|D) = P(A|D)P(B|D) for any D

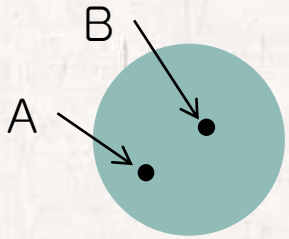
U={1,2,3,4,5,6}, A={2,4,6}, B={3,6}, C={4,5,6}

Gaussian Distribution

- Mahalanobis Distance



Data distribution



Data distribution

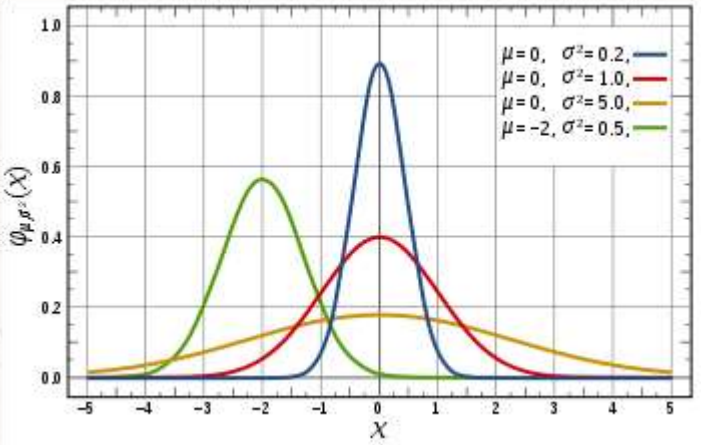
$$p_1 = \left(x_1^1, x_2^1, \cdots, x_k^1\right), p_2 = \left(x_1^2, x_2^2, \cdots, x_k^2\right)$$

$$D_M(p_1, p_2) = \sqrt{(p_1 - p_2)\Sigma^{-1}(p_1 - p_2)^T}$$

Gaussian Distribution

- Gaussian Distribution (Normal Distribution)

$$X \sim N(\mu, \sigma)$$



Mahalanobis distance between x and μ

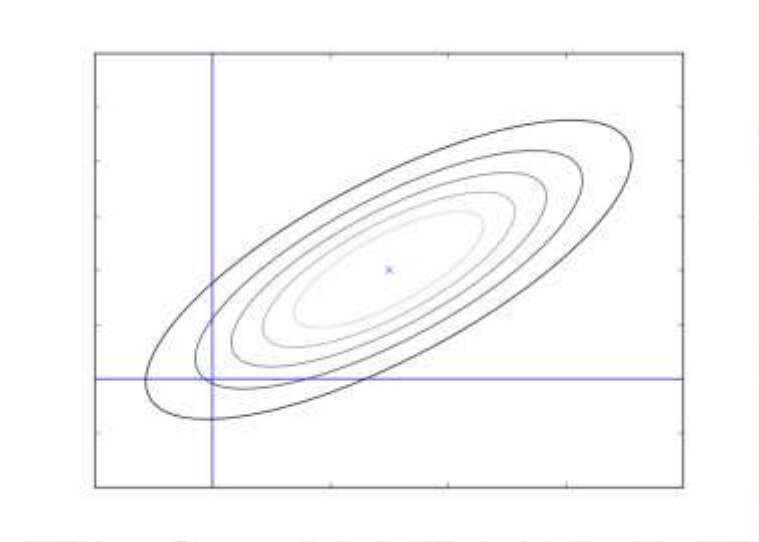
$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2} \cdot \frac{(x-\mu)^2}{\sigma^2} \right)$$

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{D_M(x,\mu)^2}{2} \right)$$

Gaussian Distribution

- Gaussian Distribution (Normal Distribution)

$$X \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$



$$f(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{m}{2}} \|\boldsymbol{\Sigma}\|^{\frac{1}{2}}} \exp\left(-\frac{D_M(\mathbf{x}, \boldsymbol{\mu})^2}{2} \right)$$

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{m}{2}} \|\boldsymbol{\Sigma}\|^{\frac{1}{2}}} \exp\left(-\frac{(\mathbf{x}-\boldsymbol{\mu})\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})^T}{2} \right)$$

Gaussian Distribution

- Covariance

$$\mathrm{cov}(X_1, X_2) = \sigma^2_{X_1,X_2} = E(X_1 X_2) - \mu_{X_1}\mu_{X_2}$$

$$= E\left((X_1 - \mu_{X_1})(X_2 - \mu_{X_2})\right) = \frac{1}{n}\sum_{i=1}^{n}(x_1^i - \mu_{X_1})(x_2^i - \mu_{X_2})$$

- Covariance Matrix
  - *k* variables are observed together

$$(x_1^1, x_2^1, \cdots, x_k^1), (x_1^2, x_2^2, \cdots, x_k^2), \cdots, (x_1^n, x_2^n, \cdots, x_k^n)$$

$$\Sigma = \begin{pmatrix} \sigma^2_{1,1} & \sigma^2_{1,2} & \cdots & \sigma^2_{1,k} \\ \sigma^2_{2,1} & \sigma^2_{2,2} & \cdots & \sigma^2_{2,k} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma^2_{k,1} & \sigma^2_{k,2} & \cdots & \sigma^2_{k,k} \end{pmatrix}$$

where $\sigma_{i,j} = \sigma_{X_i,X_j}$

Symetric because $\sigma_{i,j} = \sigma_{j,i}$

Gaussian Distribution

- Covariance

$$(0,0),(1,2),(2,4),(0,2),(1,0),(1,4),(2,2)$$

$$\mu_{X_1} = (0+1+2+0+1+1+2)/7 = 1$$
$$\mu_{X_2} = (0+2+4+2+0+4+2)/7 = 2$$
$$\mu_{X_1X_1} = (0+1+4+0+1+1+4)/7 = 11/7$$
$$\mu_{X_2X_2} = (0+4+16+4+0+16+4)/7 = 44/7$$
$$\mu_{X_1X_2} = (0+2+8+0+0+4+4)/7 = 18/7$$
$$\sigma^2_{X_1,X_1} = \mu_{X_1X_1} - \mu_{X_1}\mu_{X_1} = 11/7 - 1 = 4/7$$
$$\sigma^2_{X_2,X_2} = \mu_{X_2X_2} - \mu_{X_2}\mu_{X_2} = 44/7 - 4 = 16/7$$
$$\sigma^2_{X_1,X_2} = \mu_{X_1X_2} - \mu_{X_1}\mu_{X_2} = 18/7 - 2 = 4/7$$

$$\Sigma = \begin{pmatrix} 4/7, & 4/7 \\ 4/7, & 16/7 \end{pmatrix}$$

Gaussian Distribution

- Covariance Matrix

$$\Sigma = \begin{pmatrix} \sigma^2 & 0 & 0 & \cdots & 0 & 0 \\ 0 & \sigma^2 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \sigma^2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \sigma^2 & 0 \\ 0 & 0 & 0 & \cdots & 0 & \sigma^2 \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \sigma^2_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & \sigma^2_2 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \sigma^2_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \sigma^2_{m-1} & 0 \\ 0 & 0 & 0 & \cdots & 0 & \sigma^2_m \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \sigma^2_1 & \sigma_{12} & \cdots & \sigma_{1m} \\ \sigma_{12} & \sigma^2_2 & \cdots & \sigma_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1m} & \sigma_{2m} & \cdots & \sigma^2_m \end{pmatrix}$$

